STA 3381-04 | Probability & Statistics

Course Information

Fall 2026

Mon/Wed/Fri 9:05-9:55 AM

Dr. Jeanne Hill

Office	MMSci 318
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Phone	254-710-4081

Office Hours:

Monday	Tuesday	Wednesday	Thursday	Friday
1:30-3:30 PM	1:30-3:30 PM	1:30-2:30 PM	1:30-3:30 PM	1:30-2:30 PM

*Other times are available by appointment, and if my door is open, feel free to stop by!

Textbook: *Probability & Statistics for Engineering and the Sciences, 9th Edition* by Jay Devore.

Course Description

This course introduces the principles of probability and statistics with applications in engineering and the physical sciences. Students will develop the ability to model uncertainty using probability, analyze and interpret data, and apply statistical reasoning to make informed decisions. Emphasis is placed on problem-solving, critical thinking, and the ethical use of data in addressing real-world challenges.

Consistent with Baylor University's mission of educating students for worldwide leadership and service, this course seeks to develop both analytical skills and intellectual integrity. Students will learn to use statistical methods responsibly, evaluate evidence thoughtfully, and communicate quantitative information effectively in ways that contribute to their professions and communities.

Course Objectives

By the end of this course, students will be able to:

- Understand and apply fundamental concepts of probability theory and probability distributions.
- Analyze data using appropriate descriptive and inferential statistical methods.

- Construct and interpret confidence intervals for population parameters.
- Formulate and conduct hypothesis tests and draw appropriate conclusions from statistical evidence.
- Apply statistical reasoning to real-world problems in engineering, science, and related fields.
- Communicate statistical findings clearly and ethically.
- Recognize the role of quantitative evidence in informed decision-making and responsible leadership.

Topics Covered

Selected topics from Chapters 2–9 and Chapter 12 of Devore's textbook:

- **Chapter 2:** Probability
- **Chapter 3:** Discrete Random Variables
- **Chapter 4:** Continuous Random Variables
- **Chapter 5:** Joint Probability Distributions and Sampling Distributions
- **Chapter 6:** Point Estimation
- **Chapter 7:** Interval Estimation
- **Chapter 8:** One-Sample Hypothesis Testing
- **Chapter 9:** Inferences Based on Two Samples
- **Chapter 12:** Simple Linear Regression

Course Materials

Textbook

Probability and Statistics for Engineering and the Sciences (9th Edition), Jay Devore.

Canvas & Course Materials
Lecture materials, chapter outline notes, assignments, and course announcements will be posted on Canvas. We will work through the outline notes together during class. Students may take notes digitally using Notability or a similar app, or print the notes for use in class. Students are responsible for checking Canvas and their Baylor email regularly for course updates. If printing notes, using a binder to stay organized is recommended.

Calculator

Students should bring a scientific calculator capable of performing operations such as arithmetic computations, powers, exponentials, logarithms, and combinations. Graphing and statistical calculators are permitted but not required. Only calculators permitted on the SAT and ACT may be used during exams.

Software

This course incorporates statistical computing as an important component of data analysis. Students will use statistical software to complete selected assignments and interpret computer output.

Software used in the course may include:

- R: https://www.r-project.org/
- RStudio (Posit): https://posit.co/download/rstudio-desktop/

Interpretation of statistical software output will be an important component of homework, quizzes, and exams.

Artificial Intelligence (AI)

Unless otherwise specified by the instructor, all assignments submitted for credit must represent your own work and should not be produced, in whole or in part, using artificial intelligence tools (including but not limited to ChatGPT, Grammarly, Copilot, or similar technologies).

Unauthorized use of artificial intelligence constitutes a violation of Baylor University's Honor Code and academic integrity policies. At times, specific AI tools may be permitted or required for instructional purposes. In such cases, clear guidelines regarding appropriate and ethical use will be provided.

As members of the Baylor community, students are expected to demonstrate honesty, integrity, and personal responsibility in all academic work. The ethical use of emerging technologies is consistent with Baylor's commitment to developing principled leaders who use knowledge and innovation responsibly in service to others.

Prerequisites

Calculus II or higher.

Students should be familiar with:

- The Fundamental Theorem of Calculus
- Integration techniques including substitution and integration by parts
- Differentiation using product, quotient, and chain rules

Homework and Quizzes

Practice problems from the textbook will be assigned regularly to reinforce concepts discussed in class. These assignments are intended to support student learning and generally will not be collected unless otherwise specified.

Additional graded homework assignments will be administered throughout the semester. Weekly quizzes will be given to assess understanding of recent material and encourage consistent engagement with the course content. The lowest quiz grade will be dropped when computing the final quiz average.

Success in statistics requires consistent practice. Students are encouraged to approach assignments not merely as requirements but as opportunities to develop habits of disciplined inquiry, problem-solving, and lifelong learning.

Exams

There will be three comprehensive examinations during the semester.

Exams are closed-book; however, students may bring:

- One 8.5" × 11" sheet of notes
- One approved calculator

To maintain the integrity of the testing environment, all phones, smartwatches, smart glasses (e.g., Ray-Ban Meta, Google Glass), earbuds/headphones, and other electronic devices must be turned off and placed along the walls of the classroom before the exam begins. Any such item found on a student's person during an exam will result in a grade of zero for that exam.

Students approved for testing accommodations through OALA are responsible for reserving a testing space with OALA at least one week prior to each exam.

The comprehensive final examination is scheduled for Monday, December 14, from 9:00–11:00 AM.

Additional exam dates are tentatively scheduled for:

- Exam 1: September 25
- Exam 2: November 6

Students who miss an exam due to a documented university-approved absence or other excused reason must contact the instructor as soon as possible and provide appropriate documentation. A make-up exam may be administered at the instructor's discretion.

Grading

Component	Weight
Homework	10%
Quizzes	10%
Lowest Exam Score	20%
Median Exam Score	25%
Highest Exam Score	35%

Grading Scale

Percentage	Grade
[90, 100]	A
[87, 90)	B+
[80, 87)	B
[77, 80)	C+
[70, 77)	C
[67, 70)	D+
[60, 67)	D
[0, 60)	F

The grading scale will not be raised but may be lowered if deemed appropriate. Minus grades will only be assigned if the grading scale is lowered.

Following mathematical convention, a square bracket **[ ]** indicates that an endpoint is included, while a parenthesis **( )** indicates that an endpoint is excluded.

Teaching Assistant & Help Session

- **Teaching Assistant:** Daniel Lim
- **Email:** *Daniel_Lim3@Baylor.edu*
- **Help Sessions:** Mondays, 7:00-8:30 PM
- **Location:** MMSci GL 61
- **First Session:** August 31
- Attend for homework help, questions, and extra practice.

Baylor Mission and Course Values

Baylor University seeks to educate men and women for worldwide leadership and service through the integration of academic excellence and Christian commitment within a caring community. In support of that mission, this course encourages students to develop both technical competence and ethical responsibility as they learn to analyze and interpret data.

Students are encouraged to:

- Pursue truth with intellectual honesty and curiosity.
- Demonstrate integrity in all academic work.
- Develop excellence in quantitative reasoning and problem-solving.
- Think critically about the interpretation and communication of data.
- Use statistical methods responsibly and ethically.
- Treat classmates, instructors, and collaborators with respect and kindness.
- Recognize that data-driven decisions can affect individuals, organizations, and society.
- Use their knowledge and talents in service to their professions, communities, and the common good.

Our classroom will be a supportive learning community where questions are welcomed, diverse perspectives are respected, and all individuals are treated with dignity and respect. Through the study of statistics, students will be better prepared to exercise leadership, make informed decisions, and serve others with competence, integrity, and humility.

Practice Problems
(Devore 9th Edition.)

Chapter	Pages	Problems
2	56-57	2,4,9
	64-66	11,12,13,14,15,19,21,25
	73-75	29,31,33, 34,36, 38
	82-85	45,46,47,49,51,55,53,59,60,63,65,67
	88-91	71,73,74,75,76,77,80,83,87
3	107-109	12,13,14,16,22,23,24
	116-117	29 a,c,d, 30,32,35,38,39,42,44a,c
	123-125	47,48,50,51,55,57,59,60,62
	130-131	68,69,71
	135-136	80,81,82,85,86,88
4	146-147	1,3,5,8
	154-156	11,12,15,19,20,21 Uniform: 14,17
	167-169	28,29,31,33,35,36,38,40,43,45,54,55
	175-176	59,60,61,64,65,67,69,70
	182-183	72,73,78,84
	192-193	87,88,94
5	210	1,3,4
	219	22,30
	236-238	46,47,49,52,54
	241-242	58,59,62,65,73
6		Supplementary HW given in Class
7	284-285	1,2,4,5,6
	294	20,21,23 (one proportion)
	302-304	28,30,33,35,37a,b, 38 a,b,c, 41
	306	42,43,44,45
8	325-326	1,2,3,4,7,9, 12,13
	333-334	15,16,18,19,21
	344-345	29,31,33,36,37, 39
	350-351	42,43,45,47
	356	53,55
9	379-380	22,23,25,26,28 ,31
	388-390	36,39,40,47
	397-398	49,50,52 (two proportions)
12		Supplementary HW given in Class
	507-508	16,17,21
	517	32
	526	50,53

Baylor University

OFFICE OF THE PROVOST

Syllabus Statements

Fall 2026

Academic Integrity

Academic dishonesty and any form of cheating involves a breach of student-teacher trust. Instances of plagiarism or any other act of academic dishonesty will be reported to Office of Academic Integrity, may be referred to the Honor Council for action, and may result in failure of the course. I expect you, as a Baylor student, to be intimately familiar with the Honor Code.

Academic and creative work submitted under your name is expected to be your own, neither composed by anyone else as a whole or in part, nor handed over to another party for complete or partial revision. Be sure to document all ideas that are not your own. Furthermore, in your completion of course work and assignments you must not use technologies which give you an unfair academic advantage for this course unless they have been explicitly permitted by me. In addition, providing course materials to other students, whether individually or generally (such as online) that would enable them to gain an unfair academic advantage is an act of academic dishonesty. If you have questions about what constitutes academic integrity in this course, please contact me.

Accommodation

Any student who needs academic accommodation related to a documented disability should inform me immediately at the beginning of the semester. You are required to obtain appropriate documentation and information regarding your accommodation from the Office of Access and Learning Accommodation (OALA). OALA is located in the basement level of the east wing of the Sid Richardson building (reachable via elevator or stairs). They can also be reached by phone (254) 710-3605 or email OALA@baylor.edu.

Artificial Intelligence (AI)

Unless I instruct you otherwise, all assignments should be your original work and should not be produced in part or in total with the assistance of artificial intelligence (for example, ChatGPT, Grammarly, or some other resource). Use of artificial intelligence without explicit permission from the instructor constitutes a violation of the Honor Code at Baylor University. I [may or will] at times use artificial intelligence and other technologies for instruction or allow or expect you to use them to complete your assignments, but in these instances I will provide clear permission and guidance for their proper use.

Attendance

Successful students maintain regular attendance in this course, arriving promptly to class and ready with the necessary materials for the day's work. My attendance policy is detailed below, and all students enrolled will be held responsible for these expectations. Per the College of Arts and Sciences, students are expected to attend at least 75% of classes. In this MWF course if you miss more than 10 classes your grade will be an F.

Classroom Community

Everyone deserves to be treated with respect and dignity in this classroom space: students, teaching assistants, instructors, and guests. Baylor's Code of Ethics (BU-PP 024) states, "Members of the Baylor University community are expected to act in a way that builds a distinctive sense of caring, kindness, mutual respect, collegiality, and fairness. The idea that the Baylor community is a family has always been vital in our historical traditions, and we here reaffirm our commitment to that idea. Out of respect for this community, we do not slander or defame." Incivility, unkindness, and aggression have no place in our classroom community.

Crisis and Emergency Numbers:

Please make a note of the following numbers for crises or emergencies:
Counseling Center Crisis Line: (254) 710-2467 (Business Hours/Non-Business Hours/Weekends)
Baylor Police Department: (254) 710-2222
MHMR Crisis Center: (254) 867-6550
MHMR 24-Hour Emergency/Crisis Number: (254) 752-3451
When home during academic breaks, when the counseling center is closed, please call your local resources and/or national hotlines:
National Hope Network Hotline: 1-800-SUICIDE (1-800-784-2433)
National Suicide Prevention Lifeline: 1-800-273-TALK (1-800-273-8255)

Sexual and Interpersonal Misconduct Policy and Civil Rights Policy

Baylor University does not tolerate unlawful harassment or discrimination on the basis of sex, gender, race, color, disability, national origin, ancestry, age, citizenship, genetic information or the refusal to submit to a genetic test, past, current, or prospective service in the uniformed services, or any other characteristic protected under applicable federal, Texas, or local law (collectively referred to as Protected Characteristics). These policies also prohibit discrimination and harassment based on pregnancy or related conditions.

If you or someone you know would like help related to an experience involving:

1. Sexual assault, sex-based harassment or discrimination, dating violence, domestic violence, stalking, sexual exploitation, or retaliation for reporting one of these types of prohibited conduct, please visit www.baylor.edu/titleix, or contact us at (254) 710-8454, or TitleIX_Coordinator@baylor.edu.
2. Harassment or discrimination (excluding those issues listed in #1) based on Protected Characteristics, please visit www.baylor.edu/civilrights, or contact us at (254) 710-8454 or Civil_Rights@baylor.edu.

For pregnancy or related conditions, the University will provide reasonable modifications to policies, practices, or procedures as necessary to ensure equal access to educational programs and activities. Individuals who need to request reasonable modifications should contact the Office of Access and Learning Accommodation. More information on pregnancy and parenting resources can be found here.

First-Generation Students

Baylor University defines a first-generation college student as a student whose parents did not complete a four-year college degree. The First in Line program is a support office for first-generation college students to utilize if they have any questions or concerns. Please check out First in Line, visit us in Sid Richardson, or email firstinline@baylor.edu.

Military-Connected Students

Veterans, active-duty military personnel, and dependents are encouraged to connect with the VETS program, a space dedicated to supporting our military-connected students. Please communicate, in advance, if possible, any special circumstances (e.g., upcoming deployment, drill requirements, disability accommodations).

Mission

The mission of Baylor University is to educate men and women for worldwide leadership and service by integrating academic excellence and Christian commitment within a caring community.

Office Hours

One of the best ways to take full advantage of learning in my course is by coming to my posted office hours. I look forward to guiding you in your academic pursuits. Take advantage of the hours listed or email me for an appointment.

Success Center

I believe every student who has been admitted to Baylor can be successful, and I want to partner with you to help you thrive academically. Be sure to take advantage of the many resources available for academic success, including coming to see me during my office hours. Students who regularly utilize the great resources in the Paul L. Foster Success Center such as tutoring, Learning Lab, and Academic Mentoring are among my most successful students. If your academic performance in this class is substandard, I will submit an Academic Progress Report to the Success Center so that the team of coordinated care professionals can ensure that you get the help you need.

Undergraduate Research Opportunities – ForagerOne

Baylor University is strongly committed to providing meaningful research opportunities for undergraduates. To make connecting with faculty mentors easier, Baylor uses ForagerOne, an innovative platform designed to streamline and enhance the student-faculty match process.

Students interested in research should create a ForagerOne profile, which allows you to explore research opportunities across disciplines, connect directly with faculty mentors, and showcase your interests and skills.

Religious Observances

I am committed to providing a caring environment in which students from diverse religious backgrounds can succeed. Please review the course requirements to identify assignments, exams, or attendance expectations that conflict with religious observance and related cultural practices. I ask that you communicate any conflicts to me within the first week of the semester to allow time for us to discuss the conflict and determine any fair and reasonable adjustments to the schedule and/or assignments. Keep in mind that some changes may simply not be possible for practical reasons of scheduling, but I welcome the opportunity to discuss this with you.

Student Wellbeing

I recognize that students face many struggles during their Baylor career. The Department of CARE Team Services is a group of dedicated and caring case managers who work with students facing mental health issues, financial struggles, and anything else affecting a student's ability to be successful in the classroom. You can find them on the second floor of the Student Life Center, suite 207. Contact Information: (254) 710-2100 ; CareTeam@baylor.edu.

Health Care

Baylor University is strongly committed to addressing the physical well-being and mental health of students by providing access to on-campus healthcare resources. Baylor Health Services includes Primary Care, Psychiatry, Physical Therapy and Pharmacy and is staffed with fully certified and licensed physicians and nurse practitioners, as well as nurses and administrative staff. Appointments may be made by calling their main number or by logging into the health portal located on their website. Contact Information: (254) 710-1010; Health_Services@baylor.edu.

Mental Health

You also have access to mental health services through the The Counseling Center. Connect with them for an initial appointment, goal-directed individual therapy, group therapy, nutritional counseling, and case management services. You can schedule your 30-minute initial appointment online through the health portal or by calling (254) 710-2467. If you are experiencing suicidal thoughts, call or text #988 anytime day or night for support.

Telehealth

All Baylor students also have access to medical and counseling services via telehealth in addition to the on-campus services. Medical services include virtual urgent care, psychiatry, and nutrition counseling. Counseling services include 24/7/365 in-the-

moment support and ongoing mental health counseling. Access to care is available in the evenings, during weekends, and when the University is closed. All services are free and unlimited to Baylor students. Please visit https://baylor.academiclivecare.com/ for more information.

Addiction

The Beauchamp Addiction Recovery Center (BARC) supports students in recovery from substance and behavioral addictions through an all-encompassing level of support approach that includes one-on-one mentorship, support groups, and social events open to all Baylor students. Located in the East Village Residential Community (bottom floor of Teal Residential College). Contact Information: (254) 710-7092; BARC@baylor.edu

Food Insecurity

At Baylor, we want all students to have access to food resources that will support their holistic well-being and success. If you or someone you know experiences food insecurity at any time, you can find information on campus and community food resources by visiting The Store. You can also contact Store staff at (254) 710-4931. For additional basic needs assistance, please reach out to CASE or the Care Team.

Spirituality

The Office of Spiritual Life offers programs, persons, and resources to nurture theological depth, spiritual wholeness, and missional living. Located on the corner of 5th and Speight Street in the Bobo Spiritual Life Center. Contact Information: (254) 710-3517; Spiritual_Life@baylor.edu.